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DEPARTMENT OF COMPUTER SCIENCE AND ENGINEERING

Adaptive Soil Fertility Monitoring System Using Edge Intelligence

IoT and Embedded Computing – CS344AI

Experiential Learning

REPORT

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Certificate

Certified that the **IoT and Embedded Computing** Experiential Learning titled ***"Adaptive Soil Fertility Monitoring Using Edge Intelligence"*** is carried out by in partial fulfilment for the requirement of degree of **Bachelor of Engineering in Computer Science and Engineering** of the Visvesvaraya Technological University, Belagavi during the year 2026- 2027. It is certified that all corrections/suggestions indicated for the Internal Assessment have been incorporated in the report.

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1 Abstract

This project presents the design and implementation of an Adaptive Soil Fertility Monitoring System using Edge Intelligence, aimed at enabling real-time, data-driven agricultural decision-making. The system utilizes an ESP32-based embedded platform integrated with an RS485 soil sensor module to continuously monitor key soil parameters, including nitrogen (N), phosphorus (P), potassium (K), moisture, temperature, and electrical conductivity. Unlike conventional soil monitoring solutions that rely on static threshold-based evaluation or cloud-dependent analysis, the proposed system performs all critical computations locally on the edge device, ensuring low latency, reduced communication overhead, and independence from continuous internet connectivity.

The system incorporates a lightweight Cumulative Sum (CUSUM) drift detection algorithm to identify temporal changes in soil fertility trends, enabling early detection of nutrient depletion before critical thresholds are reached. Additionally, an adaptive site calibration mechanism is implemented to dynamically establish baseline fertility levels specific to each deployment location, thereby improving classification accuracy across diverse soil types. A machine learning model, trained offline using scikit-learn and deployed on the ESP32 using m2cgen, classifies soil fertility into distinct categories based on both raw sensor inputs and derived features such as drift score and time since fertilization. Furthermore, the system includes a fertilization event logging feature that tracks post-application nutrient response and evaluates fertilizer effectiveness in real time.

The overall architecture follows a modular pipeline consisting of sensor data acquisition, feature extraction, drift analysis, adaptive calibration, and on-device inference, with results displayed locally via an OLED interface and transmitted selectively to a monitoring dashboard using MQTT. By combining edge computing, statistical drift detection, and embedded machine learning, the proposed system moves beyond conventional monitoring to provide a context-aware and adaptive soil intelligence solution suitable for precision agriculture applications.

2 Problem Statement

Efficient soil fertility management is a critical requirement in modern agriculture to ensure optimal crop yield and sustainable resource utilization. In many practical scenarios, farmers rely on intuition, periodic observation, or infrequent laboratory testing to assess soil health. These approaches are often inadequate, as they fail to provide continuous insights into soil nutrient dynamics and do not capture temporal variations in soil conditions. As a result, fertilizer application decisions are frequently suboptimal, leading to over-fertilization, under-fertilization, increased input costs, and potential environmental degradation.

2.1 Limitations of Existing Soil Monitoring Systems

Existing soil monitoring solutions exhibit several limitations:

- Most systems provide only instantaneous sensor readings without considering temporal trends or patterns in soil fertility.
- Fixed threshold-based classification fails to account for variability in soil characteristics across different locations, resulting in inaccurate assessments.
- Conventional systems lack the capability to detect early-stage nutrient depletion, as alerts are triggered only after critical thresholds are crossed.
- Absence of mechanisms to evaluate the effectiveness of fertilizer application, leaving farmers unaware of whether corrective actions have produced the desired outcome.
- Heavy reliance on cloud-based processing in some systems leads to latency, increased communication overhead, and dependency on continuous internet connectivity.

2.2 Challenges

Designing an intelligent soil fertility monitoring system involves several challenges:

- Accurate acquisition and interpretation of multiple soil parameters such as NPK, moisture, temperature, and electrical conductivity.
- Detection of gradual changes and trends in soil fertility rather than relying solely on static values.
- Adapting the system to site-specific soil conditions without requiring manual calibration or expert intervention.
- Implementing computationally efficient algorithms suitable for resource-constrained edge devices such as ESP32.
- Integrating machine learning models into embedded systems while ensuring low latency and minimal resource usage.
- Providing meaningful feedback regarding fertilizer effectiveness based on observed changes in soil parameters over time.

2.3 Proposed Solution

To address the above limitations and challenges, this project proposes an Adaptive Soil Fertility Monitoring System using Edge Intelligence. The proposed solution:

- Continuously monitors key soil parameters using an RS485-based sensor interfaced with an ESP32 microcontroller.
- Implements a CUSUM-based drift detection algorithm to identify early-stage changes in soil fertility trends.

- Employs an adaptive calibration mechanism to dynamically establish baseline conditions specific to each deployment environment.
- Integrates a machine learning classifier deployed on-device to categorize soil fertility levels based on both raw and derived features.
- Incorporates a fertilization event logging mechanism to track nutrient response and evaluate the effectiveness of applied fertilizers.
- Utilizes a lightweight communication framework to transmit only processed insights and alerts, reducing dependency on cloud infrastructure.

3 Literature Review

Recent advancements in precision agriculture have emphasized the importance of integrating sensing technologies, data analytics, and intelligent decision-making systems to improve crop productivity and resource efficiency. Several studies have explored the use of Internet of Things (IoT) and machine learning techniques for soil monitoring and agricultural optimization.

AI-driven smart agriculture systems have been proposed to combine soil analysis, irrigation management, and fertilizer recommendations into a unified framework. These systems demonstrate the potential of integrating multiple data sources to enhance decision-making; however, they often rely heavily on cloud-based computation, which may introduce latency and connectivity constraints [1]. Similarly, IoT-based soil monitoring systems incorporating deep learning techniques have shown improved accuracy in predicting soil conditions and optimizing crop yield. Despite their effectiveness, such systems typically depend on large datasets and centralized processing, limiting their applicability in resource-constrained environments [2].

Drift detection techniques, particularly those based on the Cumulative Sum (CUSUM) method, have been widely studied for identifying gradual changes in data streams. Research on optimizing CUSUM parameters highlights its effectiveness in detecting subtle variations in system behavior over time [3]. Adaptive CUSUM models further enhance detection capability by dynamically adjusting thresholds based on data characteristics, making them suitable for applications involving non-stationary environments such as soil monitoring [4]. These methods provide a strong foundation for incorporating temporal analysis into agricultural systems.

Sensor-based soil monitoring has also been extensively investigated. Studies on calibration and performance evaluation of soil sensors emphasize the importance of accurate and reliable measurements, especially under varying environmental conditions [5]. IoT systems designed for soil nutrient monitoring using MQTT communication demonstrate efficient data transmission and real-time monitoring capabilities, but often focus primarily on data collection rather than intelligent interpretation [6].

Machine learning techniques have been increasingly applied to soil nutrient monitoring and crop recommendation systems. Such approaches enable classification and prediction based on historical and real-time data, improving the precision of agricultural practices [7]. However, many implementations rely on cloud-based processing and do not fully exploit the potential of edge computing for real-time inference.

From a hardware perspective, RS485-based soil sensors have gained popularity due to their robustness, long communication range, and reliable digital output, making them suitable for field deployment in IoT systems [8]. These sensors simplify integration with microcontrollers and reduce the need for complex signal conditioning.

Broader research in agriculture highlights the growing role of deep learning and IoT platforms in enabling smart farming solutions. Surveys on deep learning applications in agriculture indicate significant improvements in tasks such as crop monitoring, disease detection, and soil analysis, while also identifying challenges related to data availability and computational requirements [9]. IoT platforms designed for smart farming demonstrate the feasibility of large-scale deployments, but also reveal practical challenges such as energy efficiency, scalability, and system integration [10].

From the above studies, it is evident that while existing systems effectively address individual aspects such as sensing, data transmission, or prediction, there is a lack of integrated solutions that combine real-time monitoring, temporal drift detection, adaptive calibration, and on-device intelligence. The proposed system addresses these gaps by incorporating edge-based computation, CUSUM-based trend analysis, and machine learning inference within a unified architecture, thereby providing a more comprehensive and practical solution for soil fertility monitoring.

4 Objectives

The primary objectives of this project are as follows:

- To design and implement an Adaptive Soil Fertility Monitoring System using Edge Intelligence for real-time analysis of soil health parameters.
- To develop a system capable of continuously monitoring key soil attributes, including nitrogen (N), phosphorus (P), potassium (K), moisture, temperature, and electrical conductivity, using embedded sensor modules.
- To implement a lightweight drift detection mechanism based on the CUSUM algorithm for identifying early-stage changes in soil fertility trends.
- To design an adaptive site calibration mechanism that automatically establishes baseline soil conditions specific to each deployment location.

- To develop and deploy a machine learning model on an edge device (ESP32) for classification of soil fertility levels based on both raw sensor data and derived features.
- To incorporate a fertilization event logging system that tracks post-application nutrient response and evaluates fertilizer effectiveness.
- To design an efficient edge-based processing pipeline that minimizes reliance on cloud infrastructure and ensures low-latency decision-making.
- To implement a data visualization and monitoring framework using MQTT and dashboard tools for real-time observation of soil parameters and system alerts.
- To ensure the system is scalable, modular, and suitable for practical agricultural deployment, supporting precision farming applications.

5 Methodology

5.1 System Architecture Overview

The proposed Adaptive Soil Fertility Monitoring System is designed as a modular, edge-based architecture that performs real-time data acquisition, processing, and intelligent decision-making directly on the embedded device. The system integrates sensor interfacing, feature extraction, drift detection, adaptive calibration, and machine learning-based classification into a unified pipeline executed on an ESP32 microcontroller.

Raw Modbus RTU frames from the RS485 soil sensor are decoded into six numerical parameters — nitrogen (N), phosphorus (P), potassium (K), moisture, temperature, and electrical conductivity (EC) — at a fixed sampling interval of 30 seconds. These values are passed to the feature extraction stage, which appends two derived features: the current CUSUM statistic and elapsed time since the last logged fertilization event, producing an 8-dimensional input vector. This vector is consumed by the adaptive calibration engine, which maintains and updates the site-specific baseline, and subsequently by the on-device ML inference module. Classification results and drift alerts are dispatched concurrently to the OLED display and the MQTT publisher.

The overall system flow is illustrated below:

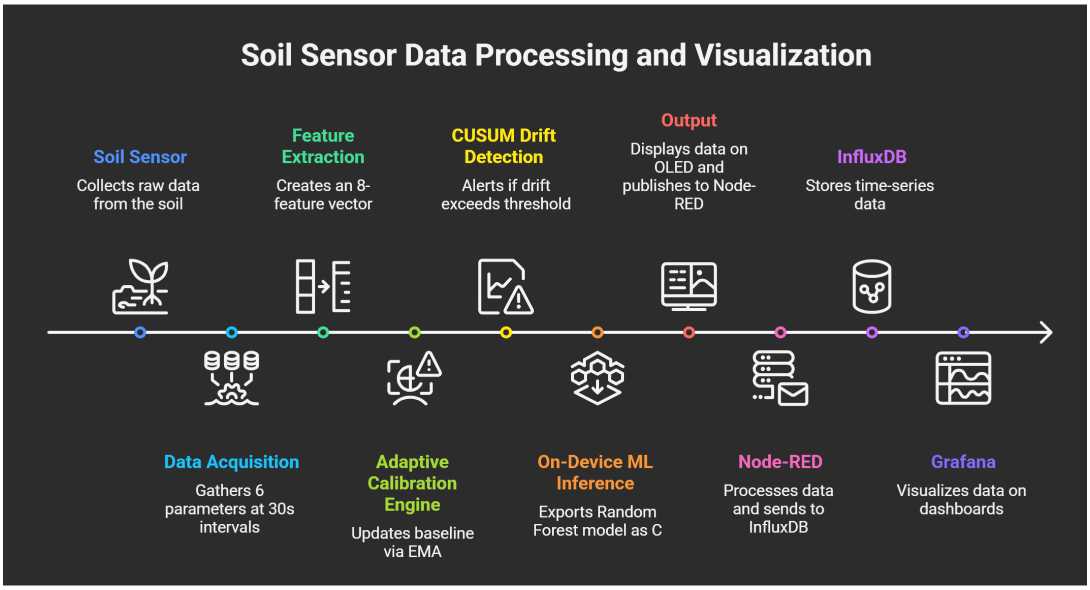

Figure 1: Overall System Architecture Flow

This architecture ensures low-latency processing, reduced communication overhead, and independence from continuous cloud connectivity.

5.2 Hardware Development Phase

The hardware subsystem is responsible for reliable acquisition of soil parameters and user interaction.

5.2.1 Soil Sensor Interface (RS485 Modbus)

A calibrated RS485-based soil sensor module is used to measure nitrogen (N), phosphorus (P), potassium (K), moisture, temperature, and electrical conductivity. Communication is established using the Modbus RTU protocol.
The sensor module performs internal signal conditioning and outputs factory-calibrated digital values, eliminating the need for external analog front-end circuitry. This design decision directly addresses the primary feasibility risk of the project, delegating measurement accuracy to a purpose-built component rather than relying on the ESP32's nonlinear ADC for raw analog soil signal processing.

5.2.2 MAX485 Communication Module

The MAX485 module is used to convert RS485 signals to TTL levels compatible with the ESP32 UART interface, enabling robust long-distance communication with minimal noise interference.The MAX485 module serves as a bidirectional signal translator between these two electrical standards.

5.2.3 ESP32 Microcontroller

The ESP32 serves as the central processing unit of the system.The sensor polling loop, CUSUM computation, adaptive calibration update, and ML inference get executed.

5.2.4 OLED Display and User Interface

An OLED display is integrated to provide real-time feedback on soil parameters, fertility classification, and system alerts. A push-button interface is included to log fertilization events manually

5.3 Data Collection Phase

- Sensor data collection is performed during a dedicated pre-deployment phase before model training begins.
- Serial communication is used to transmit real-time readings.
- A Python-based data logging system records sensor values along with timestamps.
- Data is labeled into predefined fertility classes (Depleted, Moderate, Nutrient-Rich) under controlled conditions. This dataset forms the basis for training the machine learning model.
- Data is collected under controlled conditions across three fertility classes. Samples are collected across a range of moisture levels (30–80%) and ambient temperatures (22–34°C) to ensure the model is exposed to realistic measurement variance rather than idealized uniform conditions.

5.4 Data Processing and Feature Engineering

- Raw sensor readings are preprocessed to improve model performance and system reliability.
- Normalization of features to account for varying scales.
- Temperature compensation applied to sensor readings where necessary.
- Extraction of derived features: Drift score (from CUSUM algorithm) Time since last fertilization event

These engineered features enhance the model's ability to capture temporal and contextual information.

5.5 Drift Detection Mechanism

A Cumulative Sum (CUSUM) algorithm is implemented to detect gradual changes in soil fertility trends.

- Maintains a running statistical model of recent readings.
- Computes cumulative deviations from the expected mean.
- Triggers alerts when a significant shift in nutrient levels is detected.

This enables early identification of soil degradation before critical thresholds are reached.

5.6 Adaptive Calibration Mechanism

To address variability in soil characteristics across locations, an adaptive calibration approach is employed.

- During initial deployment at a new site, the system collects the first N sensor readings and computes the site-specific baseline mean and standard deviation from these samples. These parameters are stored and used to initialize the CUSUM threshold and classification boundaries.
- Classification thresholds are dynamically adjusted relative to this baseline. The baseline is updated incrementally at each subsequent reading using an exponential moving average
- The system periodically updates calibration parameters to maintain accuracy over time.

This ensures robust performance across diverse agricultural environments.

5.7 Machine Learning Model Development

A supervised classification approach is used to categorize soil fertility into three classes: Depleted, Moderate, and Nutrient-Rich.

- Model: Random Forest Classifier / Multi-Layer Perceptron (MLP)
- Training performed using scikit-learn in a Jupyter Notebook environment
- Dataset split into training and testing subsets for validation.
- Performance evaluated using classification metrics such as accuracy and confusion matrix

The trained model is converted into a pure C implementation using m2cgen for deployment.

5.8 Model Deployment on ESP32

The exported C code of the trained model is integrated into the ESP32 firmware.

- No external ML libraries are required at runtime
- The inference pipeline within firmware executes as follows:
    1. Read raw sensor values from Modbus
    2. Apply hardcoded StandardScaler transform to each value
    3. Append current CUSUM score and time-since-fertilization
    4. Pass 8-feature array to exported C classifier function
    5. Receive integer class label (0 = Depleted, 1 = Moderate, 2 = Nutrient-Rich)
    6. Update OLED display and MQTT payload
- Enables real-time classification without cloud dependency

This approach ensures efficient utilization of limited embedded resources.

5.9 Fertilization Event Tracking

A dedicated mechanism evaluates fertilizer effectiveness by monitoring the NPK response curve following a logged application event.

- User logs fertilization events via a push-button interface
- System monitors changes in NPK values over a defined time window
- Response curves are analyzed to classify effectiveness as: Effective, Partial or Ineffective

This feature provides actionable feedback on agricultural interventions.

5.10 Communication and Visualization

Processed data and alerts are transmitted using a lightweight MQTT-based communication framework that publishes only derived insights rather than raw sensor streams, minimizing bandwidth consumption.

- MQTT protocol is used for publishing messages
- Node-RED processes incoming data streams
- InfluxDB stores time-series data
- Grafana provides real-time visualization of: Soil parameters, Drift trends, Fertility classification and Fertilization events

Only essential insights are transmitted, reducing bandwidth usage.

6 Conclusion

This project proposes the design of an Adaptive Soil Fertility Monitoring System using Edge Intelligence, aimed at improving decision-making in precision agriculture through real-time, context-aware analysis of soil conditions. The system integrates embedded sensing, statistical drift detection, adaptive calibration, and on-device machine learning to move beyond conventional threshold-based monitoring toward a more intelligent and responsive soil assessment framework.

The proposed architecture emphasizes edge-based processing, where all critical computations are performed locally on the ESP32 microcontroller. This approach reduces dependency on cloud infrastructure, minimizes latency, and enables operation in environments with limited connectivity. Key features such as temporal drift detection, site-specific calibration, and fertilization effectiveness tracking collectively enhance the system's capability to provide actionable insights rather than mere raw data.

At the current stage, the work presented is design-oriented, focusing on system architecture, algorithm selection, and implementation strategy. The methodology outlines a clear and feasible pathway for hardware integration, data collection, model training, and deployment. The modular design ensures that each component can be developed, tested, and integrated systematically during the implementation phase.

Overall, the proposed system demonstrates a scalable and practical approach to intelligent soil monitoring. Once implemented and validated, it has the potential to support efficient fertilizer usage, improve crop yield, and contribute to sustainable agricultural practices.

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